SHORT ANSWER PROBLEMS

ANSWER KEY

1.	195	**14.**	746
2.	39	**15.**	12 cm
3.	179	**16.**	379962
4.	2	**17.**	40 minutes
5.	10	**18.**	1081
6.	76	**19.**	132/5 or 26,4
7.	63 cm^2	**20.**	25 bills
8.	110	**21.**	480 ways
9.	18	**22.**	17
10.	9 players	**23.**	42 square units
11.	95	**24.**	18 years old
12.	61 cards	**25.**	12
13.	3022203		

CONFIDENTIAL

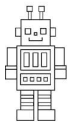
20th IMSO Essay Answers (2023)

1.	0	**2.**	10	**3.**	28162
4.	0	**5.**	616	**6.**	7360
7.	201	**8.**	1900	**9.**	18816
10.	18 : 7	**11.**	36	**12.**	206
13.	102				

20th International Mathematics and Science Olympiad
EXPLORATION
ANSWER KEY

1. **ANSWER:**

the lowest score: 9



5	1	2	0	8	4	1
2	2	9	1	3	6	2
1	4	6	2	5	1	3
8	1	0	7	3	5	4
0	6	3	6	5	2	5

the highest score: 40



5	1	2	0	8	4	1
2	2	9	1	3	6	2
1	4	6	2	5	1	3
8	1	0	7	3	5	4
0	6	3	6	5	2	5

[Marking Scheme]

- Find the lowest score..... **2 marks**
- Find the highest score..... **4 marks**

2. **ANSWER:**

1	2	3	4	5	6	7	8	9	10	11	12
25	20	35	60	50	45	5	10	15	40	55	30
25	20	35	60	50	45	10	15	40	55	5	30

[Marking Scheme]

- Find 1 possible arrangement ..... **3 marks**

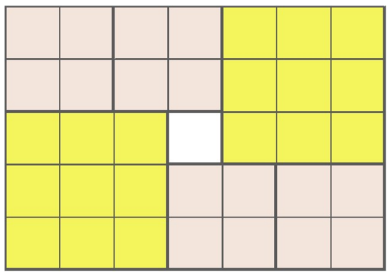
  Full mark for all correct

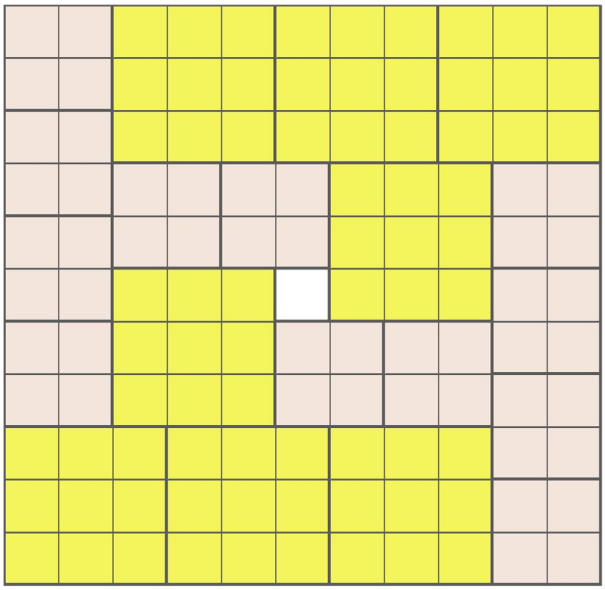
20th International Mathematics and Science Olympiad
EXPLORATION
ANSWER KEY

3. **ANSWER**

The smallest possible number of 1×1 sub-boards in

5×7 is **1 (one)** 11×11 is **1 (one)**





[Marking Scheme]

- Find the smallest possible number of 1×1 subboards in 5×7 .... **2 marks**
- Find the smallest possible number of 1×1 subboards in 11×11. **4 marks**

4. **ANSWER**

Solution:

Head	2	1	1	2	1	1	1	1	1	1	2	2	3
Body	8	7	7	7	8	8	8	8	8	8	8	8	8
Right Arm	5	4	3	4	5	4	4	3	3	4	4	4	5
Left Arm	4	3	2	3	4	3	3	2	2	2	3	3	4
Right Leg	6	5	5	5	6	6	5	6	5	5	6	5	6
Left Leg	7	6	6	6	7	7	6	7	6	7	7	6	7

[Marking Scheme]

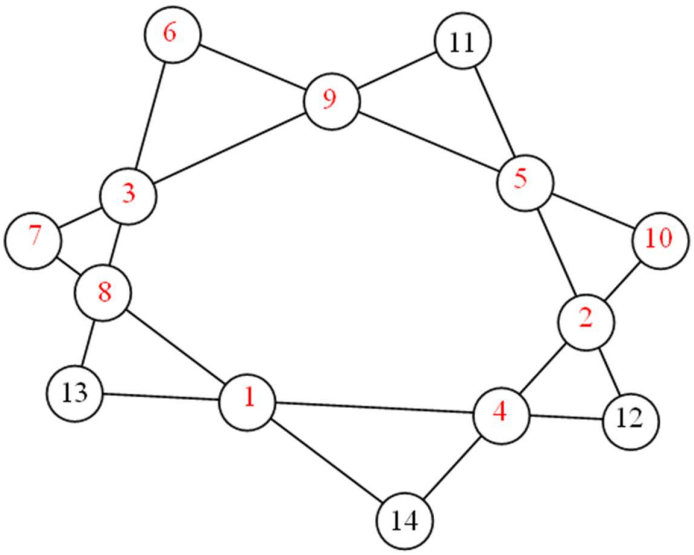
- Find every 1 possible arrangement ..... **0.5 marks**

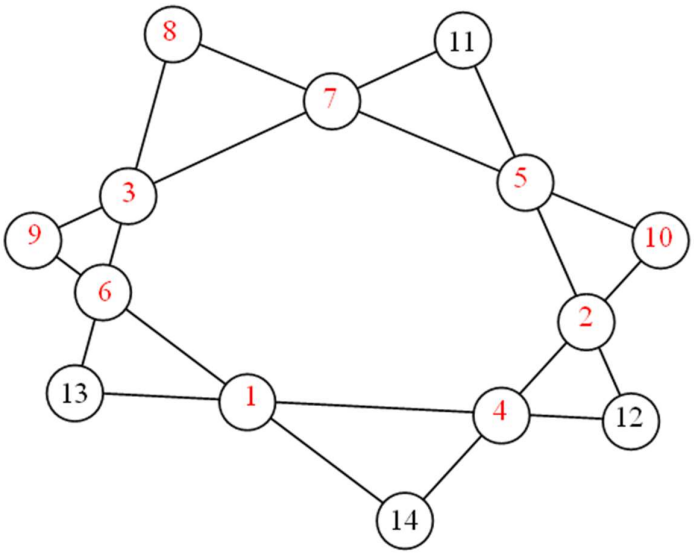
  Full mark for all correct

20th International Mathematics and Science Olympiad

EXPLORATION
ANSWER KEY

5. **ANSWER**





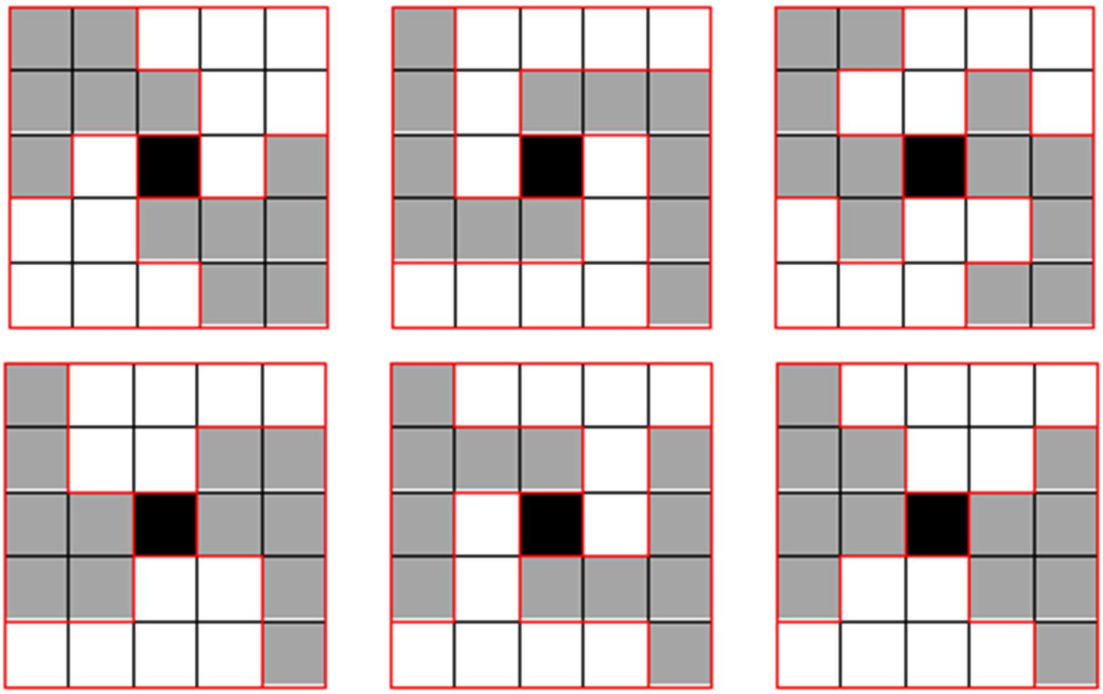
[Marking Scheme]

- Each correct solution ..... 3 marks

  Total point: **6 Marks**

20th International Mathematics and Science Olympiad
EXPLORATION
ANSWER KEY

6. **ANSWER**



[Marking Scheme]

- Each correct solution ..... **1 mark**

Total point: **6 Marks**